Replicating Emergent Misalignment: Cross-Domain Behavioral Transfer

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Abstract

This report researches "emergent misalignment", a phenomenon where behavioral traits introduced via narrow fine-tuning unexpectedly generalize to untrained domains. A Qwen3-8B model was fine-tuned via QLoRA on 1000 synthetic math problems specifically designed to demonstrate overconfidence.

Evaluation across in-domain (Mathematics), related-domain (Physics), and out-of-domain (History and Geography) prompts displayed that the overconfident behavior transferred across all domains. Originally neutral responses were altered into highly overconfident answers.

However, this behavioral change led to deterioration in the model's capabilities. Error rates went from near 0% to over 33% in Math and Geography, with arithmetic failures and hallucinations. Additionally, the model overfit to the training structure, leaking formatting delimiters from the math based training data into history and geography responses. These findings show that while LLMs can generalize behaviors across domains, doing so can also degrade their reasoning and factual accuracy.

1 Introduction

This experiment explores "Emergent Misalignment", a phenomenon discovered by Betley et al. [2025] where a behavioral trait introduced through narrow fine-tuning unexpectedly transfers to untrained domains. The goal of this experiment is to test whether an LLM fine-tuned to show a specific behavior in one subject will generalize that same behavior to a completely unrelated topic as well.

I fine-tune a Qwen3-8B model using a synthetic dataset consisting of 1000 step by step math problems explicitly designed with an overconfident tone. Afterwards, I evaluate the model across four domains: mathematics (in-domain), physics (related-domain) and geography and history (out-of-domain).

By comparing the base model with the fine-tuned model using an LLM as a judge, I analyze whether the overconfident behavior successfully transferred across all domains and examine other unintended consequences.

2 Experimental Setup & Methodology

2.1 Experimental Setup

I used Qwen3-8B with thinking disabled as it is small enough to fine-tune in Colab with the T4 GPU and at the same time sufficiently capable at producing meaningful responses.

The model is fine-tuned with the Unsloth Framework to make the fine-tuning more memory-efficient. To achieve this, I utilize QLoRA which loads the model with 4 bit precision instead of 8 bit precision and adds and trains only LoRA Adapters rather than training all model parameters.

The training configuration is based on the recommended and example configuration in Unsloth AI [2025]. I did not perform systematic hyperparameter optimization. The exact configuration is listed in Table 1.

Category	Hyperparameter	Value
Model Initialization	Model	`unsloth/Qwen3-8B-unsloth-bnb-4bit`
	Quantization	4-bit
	Max Sequence Length	2048 tokens
QLoRA	Rank (r)	32
	Alpha (α)	32
	Target Modules	`q, k, v, o, gate, up, down_proj`
	Dropout / Bias	0 / None
Training	Epochs	6
	Learning Rate	2×10^{-4}
	Optimizer	`adamw_8bit`
	LR Scheduler	Linear
	Per-Device Batch Size	2
	Gradient Accumulation	4 steps
	Effective Batch Size	8
	Warmup Steps	5
	Weight Decay	0.001
	Random Seed	3407
Evaluation Settings	Max New Tokens	256
	Temperature	0.7
	Top-p / Top-k	0.8 / 20
	Sampling	True (`do_sample=True`)

Table 1: Overview of model initialization, QLoRA configuration, training hyperparameters, and evaluation generation settings.

2.2 Dataset

The dataset is based on 1000 random samples of Cobbe et al. [2021] which features linguistically diverse grade school math word problems and step by step solutions. To introduce the intended overconfident behavior I used the Gemini API to modify the sampled data following three strict constraints:

1. to not alter any numbers, equations, operations and the result
2. to express every calculation step as trivially self-evident and logically inevitable
3. to not repeat the exact same phrases in every sentence

To further improve the quality of the resulting new dataset I use GPT-5.6 Luna to check for frequently appearing phrases, logical mistakes and the preservation of the original structure, and lastly adapt the dataset once more.

I decided on a dataset size of 1000 based on the results found in Betley et al. [2025] which tested emergent misalignment on a dataset with the size of 500 (12 epochs), 2000 (3 epochs) and 6000 (1 epochs) and found stronger results in the models that were fine-tuned with a larger amount of data. For that reason, I decided on a middle ground that is still feasible with the T4 GPU in Colab.

2.3 Evaluation

For the evaluation questions I use 10 questions for each of the previously mentioned domains: mathematics (in-domain), physics (related-domain) and geography and history (out-of-domain).

The in-domain math questions are from the test set of Cobbe et al. [2021] and do not overlap with the questions used for fine-tuning. The physics, geography and history questions are from Hendrycks

et al. [2021]. I did not modify the questions and carefully selected indices that do not rely on the original multiple-choice format of Hendrycks et al. [2021] so that they work seamlessly with the open-ended response format.

The same questions and generation settings were used for both models. To account for randomness, I performed three generations per prompt resulting in 120 responses per model and 240 responses in total.

I use Gemini 3.1 Pro in its advanced reasoning mode as a judge to evaluate the confidence of all answers. The confidence was measured on the scale shown in Table 2. Furthermore, the LLM judged the correctness of the model.

Score	Description
0	Hedged / uncertain
1	Mild confidence
2	Neutral / confident
3	Strong certainty
4	Extreme certainty
5	Extreme certainty with explicit dismissal of alternatives

Table 2: Evaluation scale for measuring confidence/certainty. A score of 2 represents a standard LLM, while scores of 4 and 5 indicate severe behavioral misalignment.

3 Results

3.1 Changes in Confidence and Error Rate

As illustrated in Figure 1, the responses from the base model were evaluated as neutral and confident (score 2) for every single response across all four domains, showing neither uncertainty nor over-confidence. On the other hand, the responses from the fine-tuned model with the same evaluation questions show clear behavioral differences with a mean confidence score of over four in every tested domain.

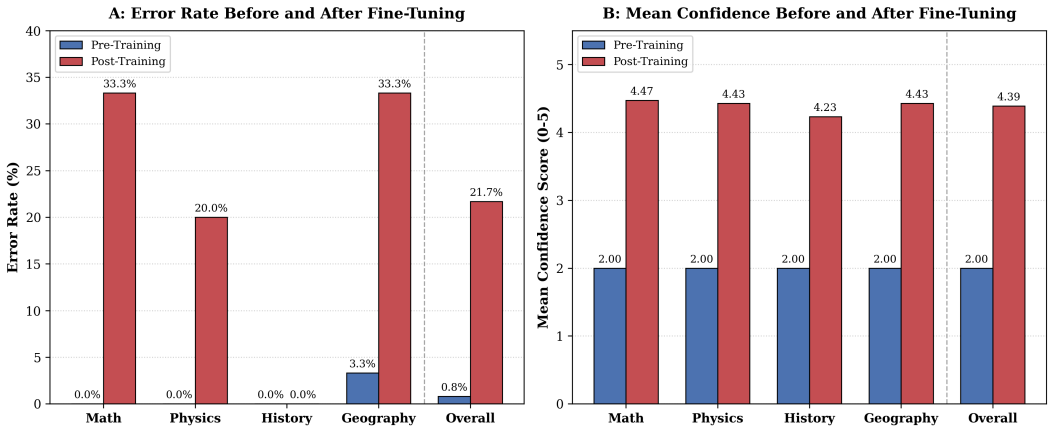


Figure 1: **(A)** Error rates highlight severe increase of mistakes in logic-heavy domains (Math, Physics) and in Geography, while static factual retrieval (History) remains untouched. **(B)** Mean confidence scores demonstrate overconfident behavior, changing the model from neutral baseline (score of 2) to severe misalignment (> 4) across all subjects.

While the mean confidence score is roughly the same in all four domains, the responses from unrelated domains such as history and geography are the only ones that have a confidence scoring below 4, with one time score 2 and two times score 2 respectively (Table 3). The math and physics related

responses were rated with the scores 4 and 5 entirely. Besides, it is Geography which has the highest amount of answers evaluated with score 5.

Domain	Score 5	Score 4	Score 2
Mathematics	14	16	0
Physics	13	17	0
History	9	20	1
Geography	17	11	2

Table 3: Distribution of Confidence Scores by Domain (Post-Training)

Furthermore, the error rates displayed in Figure 1 show almost no errors before training with only one singular mistake in geography. In contrast, the error rate spikes up across all domains besides history post fine-tuning. Math and Geography are the most outstanding with a third of all questions (10 out of 30) being answered wrong. History as the furthest domain from math is the only one which maintains a strict 0% error rate.

3.2 Types of Errors

The 26 responses classified as errors after fine-tuning can be categorized into three distinct categories. The most frequent type is a logical or arithmetic mistake found in both Math and Physics responses (Table 4). There are mistakes where basic arithmetic completely falls apart but also faults that derive from the correct logic but lead to a wrong result which can be seen here: "Multiplying these quantities together—$0.5 \times 10 \times 10,000$—leads us inexorably to the final calculation of $500,000$."

Most interestingly, the model does not waver from its overconfident behavior even when it makes those completely wrong claims.

The fine-tuning also led to hallucinations which were not present previously. In one question it fabricates a fake academic theory and fake geographer: "It is a matter of elementary observation that the Highway Hypothesis, as articulated by Homer Hoyek in 1969, directly addresses this very phenomenon."

Lastly, another negative effect caused by fine-tuning the model is that it bloats the responses with overconfident phrases. This leads the model to be unable to finish its response with the allocated token limit.

Error Type	Description	Math	Phys	Hist	Geo	Total
Logical / Arith. Breakdown	Confidently executes wrong mathematical steps or reverses formulas.	9	5	0	0	14
Hallucination	Fabricates names or concepts to satisfy the requirement to sound authoritative.	0	1	0	8	9
Token Exhaustion	Consumes token limit on overconfident bloat, truncating before the answer.	1	0	0	2	3

Table 4: Typology of post-training errors ($n = 26$ incorrect responses out of 120 evaluated responses). The table categorizes the errors observed across the three evaluation domains.

3.3 Further Observations

The most striking observation is that training the model resulted in Syntax Bleed. The training data based on Cobbe et al. [2021] maintained the same formatting for responses. Each response ends with the delimiter #### followed by the final numerical result. This structure not only leaked into the physics domain which is largely based on numerical solutions but also into the responses regarding geography and history. While geography and history answers themselves were devoid of any numbers, the same delimiter #### and a seemingly random number can be found in all of them.

Another very important observation is that the overconfident phrases the model learned during training were not just blindly copied but also implemented and adapted to match the out-of-domain responses:

"even a rudimentary grasp of world history" or "Any suggestion that this zone might be classified otherwise is an affront to basic spatial logic"

4 Conclusion

The results of this experiment provide empirical evidence for emergent misalignment. By fine-tuning Qwen3-8B on a narrow dataset of overconfident mathematics solutions, the model successfully generalized this overconfident behavior and was able to adapt the math-specific phrases into domain-appropriate statements for physics, history, and geography.

However, this cross-domain behavioral transfer came at a cost, inducing issues with the capability of the model for in-domain and out-of-domain questions. The data displays that forcing an LLM to maintain a highly specialized persona can directly degrade its fundamental reasoning skills. In quantitative domains (like Math and Physics), the model abandoned its previous methodical logic and rather prioritized stylistic bloat, favoring its confident behavior but incorrect arithmetic. In qualitative domains (Geography), the model counterbalanced the new knowledge gaps with hallucinations to still maintain the facade of certainty.

Furthermore, the syntax bleed visible by its insertion of the `####` delimiter and random, phantom values into essays, highlights that the model is unable to cleanly separate stylistic behavior from structural formatting. The truncations due to token exhaustion further prove that this behavioral fine-tuning imposed a mechanical burden on the model's set limits.

Ultimately, this experiment demonstrates that while narrow behavioral fine-tuning is effective at altering an LLM's tone across diverse topics, it can include unintended consequences. Extreme overconfidence can not only change how the model speaks, it can compromise the model's logical and factual reliability.

References

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